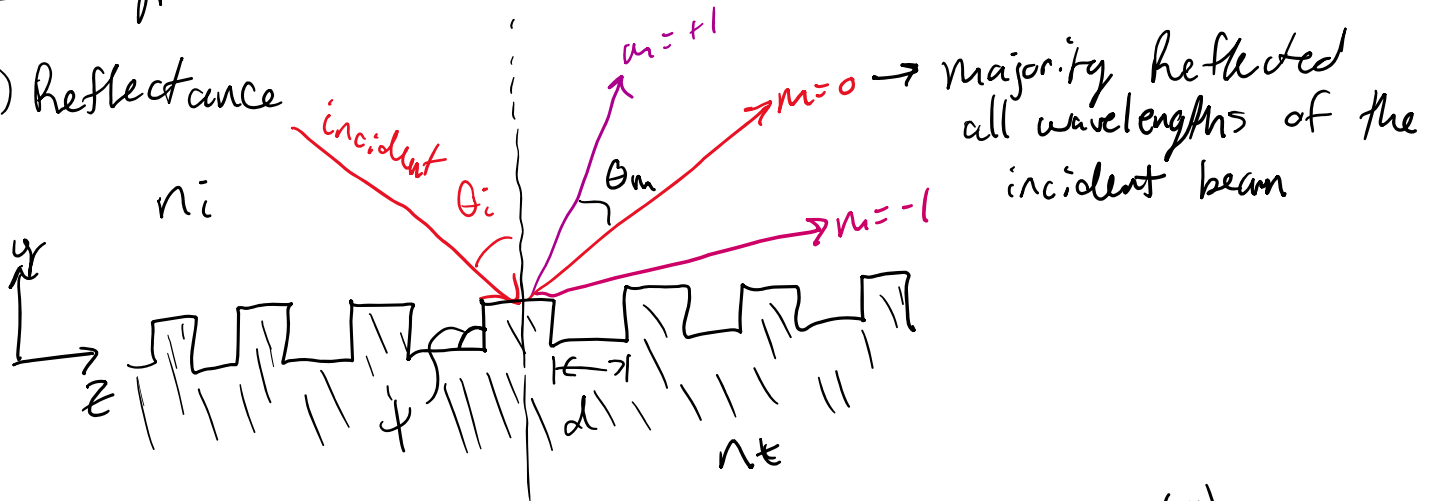


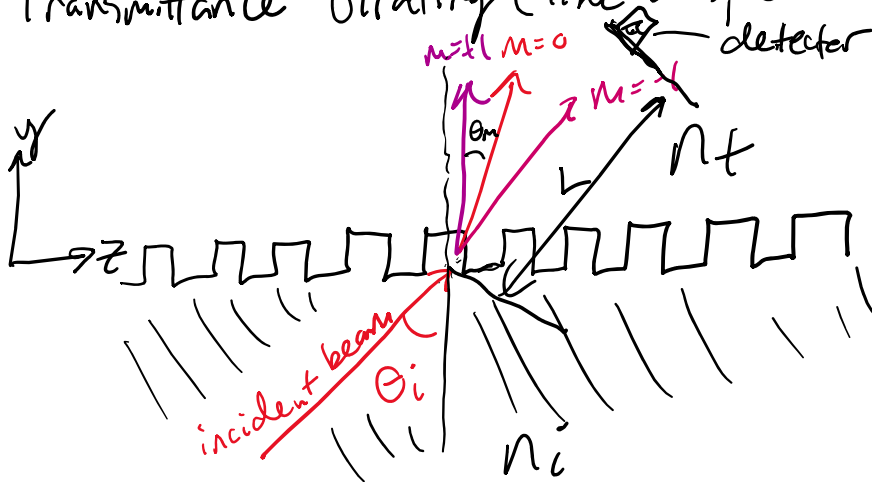
Two types of Gratings

1.) Reflectance



$$d(n_i \sin \theta_i) - n_t (\sin \theta_m) = m \lambda_0 \sin \psi$$

2.) Transmittance Grating (like a special lens)



$m = 0$ is the brightest but has all wavelengths that the incident beam has

$$d(n_i \sin \theta_i) - n_t (\sin \theta_m) = m \lambda_0 \sin \psi$$

Intensity of a Grating mode

$$I(y) = I_i \left[\frac{\sin(\frac{1}{2}(k_y)a)}{\frac{1}{2}(k_y)a} \right]^2 \left[\frac{\sin(\frac{1}{2}N k_y d)}{N \sin(\frac{1}{2}k_y d)} \right]^2$$

$N = \text{no. of slits}$
 $d = \text{periodicity of slits}$

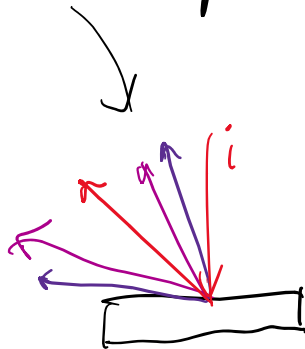
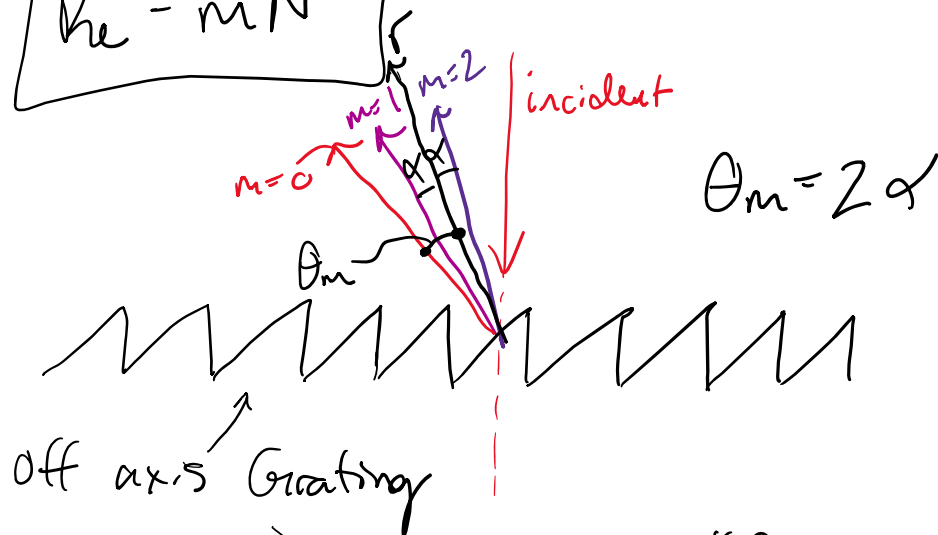
If using a detector

$L = \text{length from detector}$ $k_y = \frac{2\pi}{\lambda} \left(\frac{y}{L} \right) \approx \frac{2\pi}{\lambda} (\sin \theta_m)$

Resolvance: ability of a grating to separate out wavelengths

$$R_e = \frac{\lambda_{m=0}}{\Delta\lambda_m} = \frac{1}{\Delta\lambda_m} (\lambda_{m=2} - \lambda_{m=1})$$

$$R_e = mN$$



* Gratings are used in

- Spectroscopy

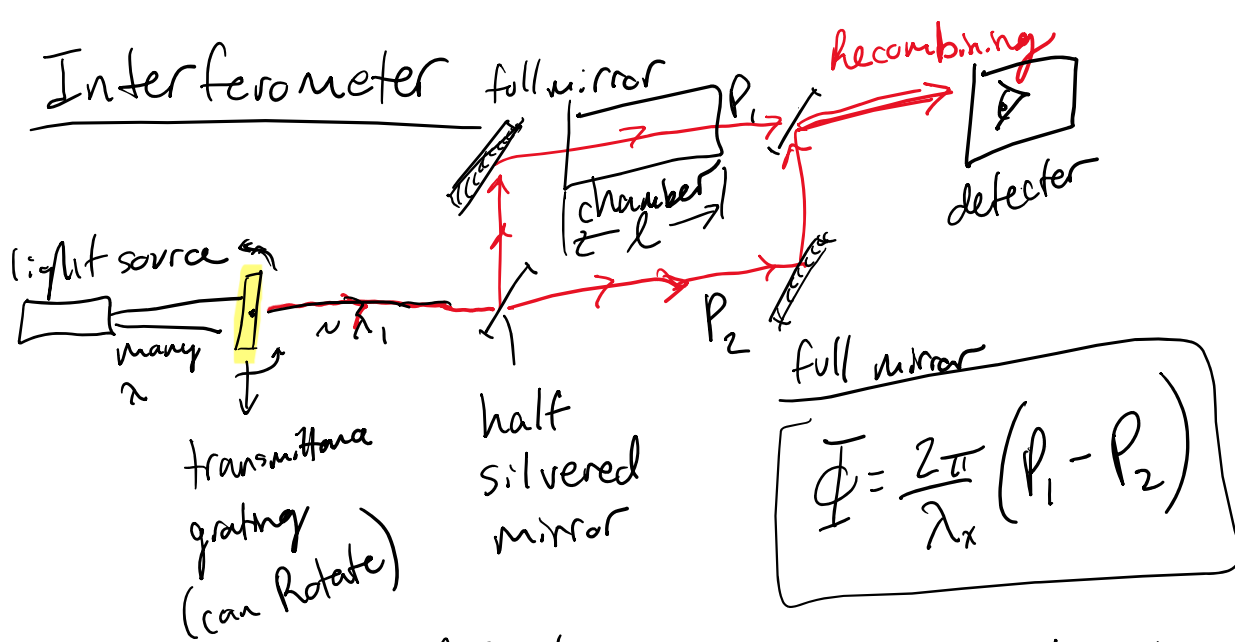
- Lasers

- Holograms

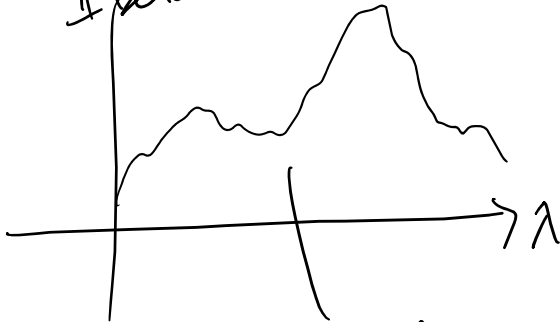
- Fresnel Grating in TVs

→
do not separate the light
Magnification (focus light)

Interferometer



look at index of refraction
Changes
Detector



spectrum of
index of refraction

P_1 is path length 1

P_2 is path length 2

$$\Phi = \frac{2\pi}{\lambda_x} (P_1 - l) + \frac{2\pi n}{\lambda_x} - \frac{2\pi P_2}{\lambda_x}$$

λ_x is the wavelength of light used.